

Recitation: Linear Functions Review Solutions

Problem 1. A small tanker truck is depositing its gas at a storage facility. The tanker is carrying 11,600 gallons of gas and is emptying its tank at a rate of 528.3 gal/min. Let $G(t)$ denote the volume of gas, in thousands, left in the tanker t minutes from now.

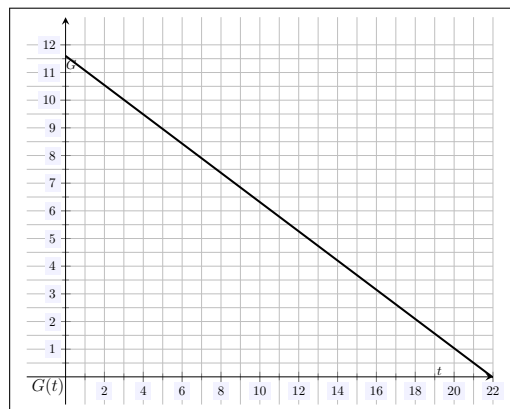
- (a) Explain why $G(t)$ is linear.
- (b) Find $G(t)$ and sketch it in the plot below.
- (c) Interpret the slope of $G(t)$.
- (d) Interpret the y -intercept for $G(t)$.
- (e) Find and interpret (if possible) the x -intercept for $G(t)$.

Solution.

- (a) The rate of change of $G(t)$, i.e. the rate at which gas is flowing out of the tanker, is constant. Therefore, $G(t)$ must be linear.
- (b) By (a), we know that $G(t)$ is linear. Therefore, $G(t) = mt + b$ for some m, b . We know m is the rate of change of $G(t)$. We are told that gas is leaving the tanker at a rate of 528.3 gal/min, i.e. 0.5283 thousands of gallons per minute. But we must then have $m = -0.5283$ because the amount of gas in the tanker is decreasing at a rate of 0.5283 thousands of gallons per minute. But then $G(t) = -0.5283t + b$. At the start, i.e. at $t = 0$, we know the tanker has 11,600 gallons of gas (11.6 thousands of gallons of gas), i.e. $G(0) = 11.6$. But then $11.6 = G(0) = -0.5283(0) + b = b$. Therefore, $G(t) = -0.5283t + 11.6$. We sketch $G(t)$ on the plot below.
- (c) We know that $m = \frac{\Delta G}{\Delta t}$, i.e. the amount of change in the gallons of gas over time. Because $m = -0.5283 < 0$, we know that the amount of gas is decreasing. As $m = -0.5283$, the slope represents the fact that every minute, 0.5283 thousands of gallons of gas is emptied from the tanker, i.e. 528.3 gallons of gas are emptied from the tanker every minute.
- (d) The y -intercept is when $t = 0$ minutes. We have $G(0) = -0.5283(0) + 11.6 = 11.6$. Therefore, the y -intercept represents the fact that at 0 minutes, the tanker is carrying 11.6 thousands of gallons of gas, i.e. there are initially 11,600 gallons of gas in the tanker.
- (e) The x -intercept occurs when $G(t) = 0$. But then we have...

$$\begin{aligned} G(t) &= 0 \\ -0.5283t + 11.6 &= 0 \\ -0.5283t &= -11.6 \\ t &\approx 21.96 \text{ minutes} \end{aligned}$$

Therefore, $G(t) = 0$ when $t \approx 21.96$, i.e. the tanker is empty after 21.96 minutes.



Problem 2. Find the equation of the line with x -intercept 5 and y -intercept -6 .

We want the equation of the line, which must have the form $y = mx + b$. We need a point and a slope. Because the x -intercept is 5, the line must contain the point $(5, 0)$. But we also know the line has y -intercept -6 , i.e. the line contains the point $(0, -6)$. Therefore, we have two points. So, we can use the two points to find the slope:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-6 - 0}{0 - 5} = \frac{-6}{-5} = \frac{6}{5}$$

So, $y = \frac{6}{5}x + b$. But we know b is the y -intercept, which is -6 . Therefore, the line must be $y = \frac{6}{5}x - 6$.

$y = \frac{6}{5}x - 6$

Problem 3. Consider the linear function $\ell(x) = \frac{3x-5}{4}$.

- (a) Find the slope and y -intercept for $\ell(x)$.
- (b) Find the x -intercept(s) of $\ell(x)$.
- (c) Is $(-1, 2)$ on the graph of $\ell(x)$?
- (d) Compute $\ell(1)$.
- (e) Determine the domain and range for $\ell(x)$.

Solution.

(a) $\ell = \frac{3x-5}{4} = \frac{3}{4}x - \frac{5}{4}$. Therefore, the slope is $m = \frac{3}{4}$ and the y -intercept is $b = -\frac{5}{4}$.

(b) The x -intercepts are where the line intersects the x -axis, where $\ell = 0$. Therefore, we have...

$$\begin{aligned}\ell &= 0 \\ \frac{3x-5}{4} &= 0 \\ 3x - 5 &= 0 \\ 3x &= 5 \\ x &= \frac{5}{3}\end{aligned}$$

Therefore, the x -intercept is $x = \frac{5}{3}$.

(c) If $x = -1$, then $\ell = \frac{3(-1)-5}{4} = \frac{-3-5}{4} = \frac{-8}{4} = -2$, i.e. $y = -2$. But then $(-1, 2)$ is on the graph. So, we know $(-1, 2)$ is not on the graph of $\ell(x)$.

(d) $\ell(1) = \frac{3(1)-5}{4} = \frac{3-5}{4} = \frac{-2}{4} = -\frac{1}{2}$.

(e) Observe we can plug in any x -value into $\ell(x)$. So, the domain of $\ell(x)$ is $\mathbb{R}$. We can also solve $\ell(x) = \#$ for any $\#$: $\frac{3x-5}{4} = \# \implies x = \frac{4\#+5}{3}$. So, the range must be all of $\mathbb{R}$. Alternatively, observe that the graph of $\ell(x)$ is a "slanted" line. So, the line "passes through" every possible y -value. Therefore, the range of $\ell(x)$ is $\mathbb{R}$.